POTHOLE SEVERITY CLASSIFIER & MAINTENANCE DECISION SUPPORT SYSTEM

Final Documentation Report

Project Report Submitted for Internship Project Submission

Submitted By

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Internship Role: Intern (AI/ML)

Project Title: Pothole Severity Classifier & Maintenance Decision Support System

Technology Stack:

Python | Machine Learning | Scikit-Learn | Support Vector Machine | XGBoost | Pandas | NumPy | Streamlit | Plotly

Project Description:

An end-to-end machine learning system developed to classify pothole severity from road inspection images, generate prioritization scores, and estimate repair costs based on scenario assumptions. The project uses machine learning models (Support Vector Machine) for severity prediction based on spatial geometry, computes an Expected Severity Priority Score, and features an interactive Streamlit dashboard with interactive Plotly metrics to visualize inspection hotspots and support municipal maintenance budget planning.

Submitted To

Company/Organization Name: Data Vidwan

Mentor/Manager Name: Meet Mistry

Date of Submission: 20/08/2026

1. Project Objective

The objective of this project is to develop a machine learning system that classifies pothole severity using historical road inspection imagery and spatial bounding box dimensions.

The system predicts pothole severity (Minor, Medium, Major), identifies high-priority inspection targets based on a probability-weighted scoring framework, and provides a fully interactive UI. This enables municipalities to optimize maintenance budgets, plan repair schedules efficiently, and distribute resources without relying on historical financial receipts.

The project includes:

• Exploratory Data Analysis (EDA)

• Data preprocessing and bounding box cleaning

• Feature engineering (relative bounding dimensions, aspect ratios)

• Multi-algorithm severity classification

• Model comparison and hyperparameter tuning

• Prioritization scoring (Critical/High/Medium/Low)

• Scenario-based maintenance cost planning (USD/INR)

• Interactive Streamlit dashboard

2. Requirement Coverage

Requirement	Status	Project File / Output
Severity Classification Model	Completed	models/final/final_model.joblib
Feature Engineering	Completed	notebooks/02_feature_engineering_and_data_splitting.ipynb
Cleaned Dataset	Completed	dataset/processed/pothole_dataset_cleaned.csv
Priority Scoring System	Completed	src/priority_analysis.py
Cost Estimation	Completed	src/cost_estimation.py

Model		
EDA Report	Completed	reports/Final_Requirement_Audit.md
Model Evaluation	Completed	reports/Municipal_Repair_Planning_Report.md
Streamlit Dashboard	Completed	dashboard/app.py
Project Documentation	Completed	Documentation/Final_Documentation_Report.docx

3. Dataset

The project uses a Pascal VOC image/object-detection dataset consisting of 717 images and 1,886 parsed pothole bounding boxes.

Features:

• bbox_width, bbox_height, bbox_area (Spatial measurements)

• rel_bbox_area (Relative dimension against original image size)

• severity (Target Variable: minor_pothole, medium_pothole, major_pothole)

4. Work Completed

Completed an end-to-end AI/ML workflow including:

• Loaded and inspected raw XML annotations.

• Performed exploratory data analysis and spatial distribution mapping.

• Extracted spatial bounding box geometry into analytical features.

• Trained 5 distinct machine learning algorithms (Random Forest, XGBoost, LightGBM, Gradient Boosting, Support Vector Machine).

• Compared model performance and evaluated metrics (Accuracy, Precision, Recall, Macro F1).

• Selected Support Vector Machine for final deployment.

• Implemented a continuous priority scoring formula based on class probabilities.

• Developed a fully functional, multi-page Streamlit dashboard with premium glassmorphism aesthetics.

• Exported processed datasets, detailed markdown reports, and trained models.

5. Methodology

5.1 Exploratory Data Analysis

Performed analysis on:

• Target class distribution (minor, medium, major)

• Dimensional correlation mapping

• Image bounding box anomalies

5.2 Feature Engineering

Created additional analytical features:

• Relative bounding box geometries (rel_bbox_width, rel_bbox_height, rel_bbox_area)

• Aspect Ratios (bbox_aspect_ratio)

• Logarithmic transformations (log_bbox_area)

6. Model Development

Multiple classification models were trained using robust Leak-Safe GroupKFold cross-validation:

• Support Vector Machine

• Gradient Boosting

• LightGBM

• Random Forest

• XGBoost

The models were evaluated using Accuracy and Macro F1 Score to account for class imbalances.

7. Model Results

Baseline validation results prior to final tuning:

Model	Macro F1 (%)	Accuracy (%)
Support Vector Machine	62.46	64.03
Gradient Boosting	58.26	59.74
LightGBM	56.66	58.42
Random Forest	56.20	58.09
XGBoost	54.71	57.10

8. Selected Model

Support Vector Machine (RBF Kernel)

Reason for selection:

The SVM significantly outperformed all tree-based ensembles on the validation groups for this specific geometric dataset, providing the highest Macro F1 score and the best robustness against overfitting.

9. Streamlit Application

The Streamlit application is implemented in:

dashboard/app.py

The dashboard contains:

• Executive Overview: Key Performance Indicators and final dataset previews.

• Severity Analysis: Visual distributions and probability heatmaps.

• Priority Framework: Histograms mapped to decision-support tiers (Critical to Low).

• Hotspot Identification: Image-level aggregation to prioritize inspection locales.

• Scenario-Based Cost Planning: Interactive USD/INR estimated budget assumptions.

10. Key Findings

• Extracted bounding box area possesses a direct correlation to labeled severity.

• Tree-based models severely overfit spatial coordinates; scaled distance-based SVMs provide superior generalization.

• Utilizing prediction probabilities generates a highly effective continuous triage score.

11. Recommendations

• Incorporate external hardware GPS modules during image capture for geographic mapping.

• Introduce LiDAR or depth-sensing arrays to compute true volumetric damage instead of 2D bounding boxes.

• Cross-reference municipal receipts to build a supervised cost regression model rather than relying on heuristic scenarios.

12. Achievements

Successfully completed:

• Data preprocessing and bounding box validation.

• Machine learning model development and comprehensive evaluation (5 models).

• Robust 4-tier decision-support priority engine.

• End-to-End multi-tabbed Streamlit Dashboard deployment with premium Plotly aesthetics.

13. Limitations

• Dataset lacks exogenous attributes such as Traffic Volume (AADT), geographic coordinates, historical repair costs, and road condition variables (IRI/PCI).

14. Final Submission Contents

• Documentation Report (.docx)

• Machine Learning Models (.joblib)

• Streamlit Dashboard (dashboard/app.py)

• Processed Datasets (outputs/)

• Markdown Reports (reports/)

• Jupyter Notebooks (notebooks/)

15. How to Run

pip install -r requirements.txt

python -m streamlit run dashboard/app.py

16. Final Conclusion

The Pothole Severity Classifier & Decision Support System successfully delivers an end-to-end machine learning solution for damage triage, severity forecasting, and transparent financial planning.

The system combines spatial AI with robust real-time heuristic modeling to help municipal managers optimize maintenance deployment and make data-driven decisions through a fully interactive, premium Streamlit dashboard.